

Copper rod at 50 Hz, driven by 1 A

Hybrid $A-\Phi$ finite-element solver, tree-cotree gauge. **The dual of the 1 V case:** the same conductor, the same mesh, the same frequency, driven with a 1 A current source instead of a 1 V terminal voltage.

```
regression_tests/03_Cylinder_1A_50Hz · 240,990 unknowns ·  
backward error 3.8e-21
```

0. Why this case exists

This is the first **current-driven** problem this solver has run. The port machinery for it had existed since the DOF map was written — a current port gets an unknown terminal potential instead of a fixed one, and its row is the terminal's current balance — but nothing had ever exercised it.

Port current self-consistency. Inject 1 A at the top cap; exactly 1 A must be collected at the bottom. Charge does not accumulate in a conductor, so any other answer means current is being created or destroyed inside the formulation. Measured: **1.000000 A**.

Does the dual agree? **R** and **L** belong to the geometry and the material, not to how the thing is driven, and the problem is linear. So driving 1 A must return the same impedance as driving 1 V:

	1 V CASE	1 A CASE
I	10180.56 A	1.000000 -0.000000 j A
V	1 V	9.796995e-05 +7.098621e-06 j V
Z	9.796995e-05 +7.098621e-06 j Ω	9.796995e-05 +7.098621e-06 j Ω
R	9.796995e-05 Ω	identical
L	22.5956 nH	identical

Identical to every digit printed.

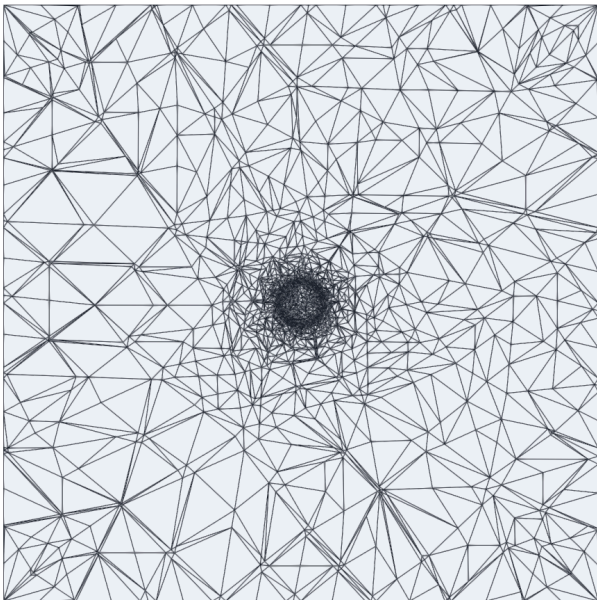
The terminal voltage is 98 microvolts, because the rod is 98 micro-ohms. Every field below is correspondingly about 10^4 times smaller than in the 1 V case — **B** peaks at $1.3e-04$ T rather than 1.34 T. That is not a red flag; it is the same physics read from the

other side, and reproducing the 1 V case's `Z` from it is a genuine test of the scaling. A formulation that quietly lost a factor somewhere would not.

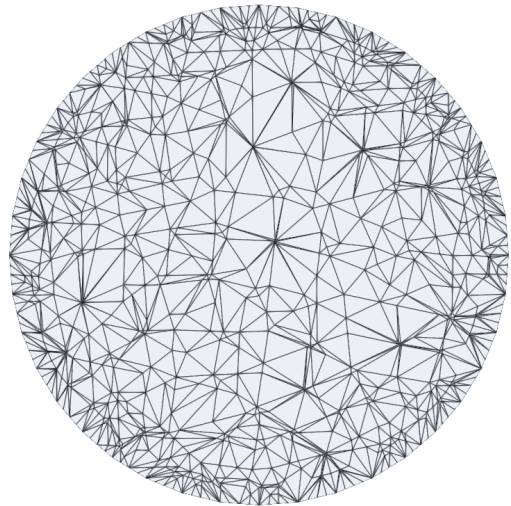
1. The case

Conductor	copper, $\sigma = 5.8\text{e}7 \text{ S/m}$, $a = 1.5 \text{ mm}$, $L = 40 \text{ mm}$, a 40-gon
Domain	40 mm cube of air, $\mathbf{n} \times \mathbf{A} = \mathbf{0}$ on the outer boundary
Excitation	1.0 A into the top cap , 0 V reference on the bottom, 50 Hz
Skin depth	$\delta = 9.346 \text{ mm}$, so $a/\delta = 0.1605$ — no skin effect
Formulation	first-order Whitney edge $\mathbf{A}$, second-order P2 nodal Φ
Mesh	the 1 V case's, referenced not copied — 18,994 nodes, 101,873 tets, Netgen-optimised

2. Mesh



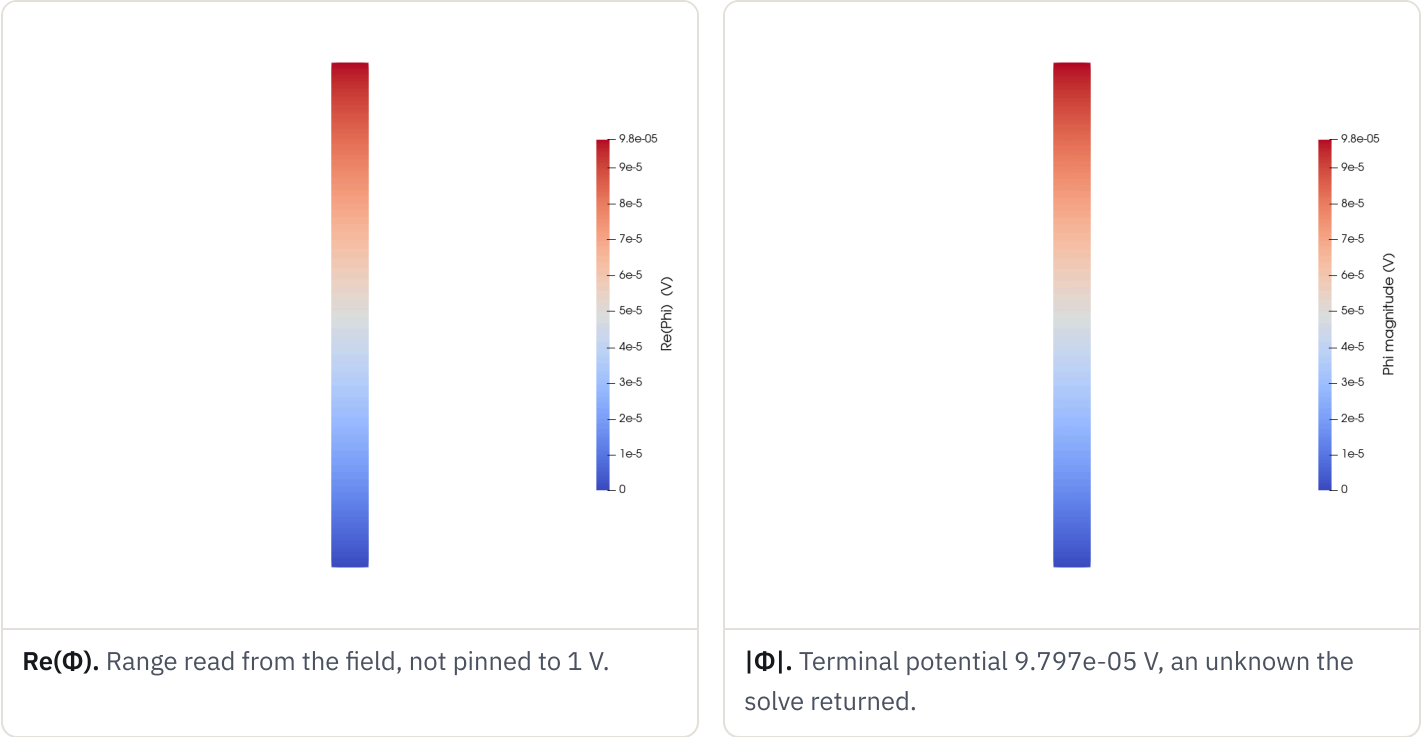
Whole domain. Graded from 0.25 mm at the conductor to 6 mm in the far air.



The wire alone. Roughly 6 elements across the radius.

Identical to the 1 V case by construction — the input file points at that case's `cylinder.msh`. "Exactly the same mesh" is the whole point of the comparison, and a copy is a thing that can drift.

3. Potential

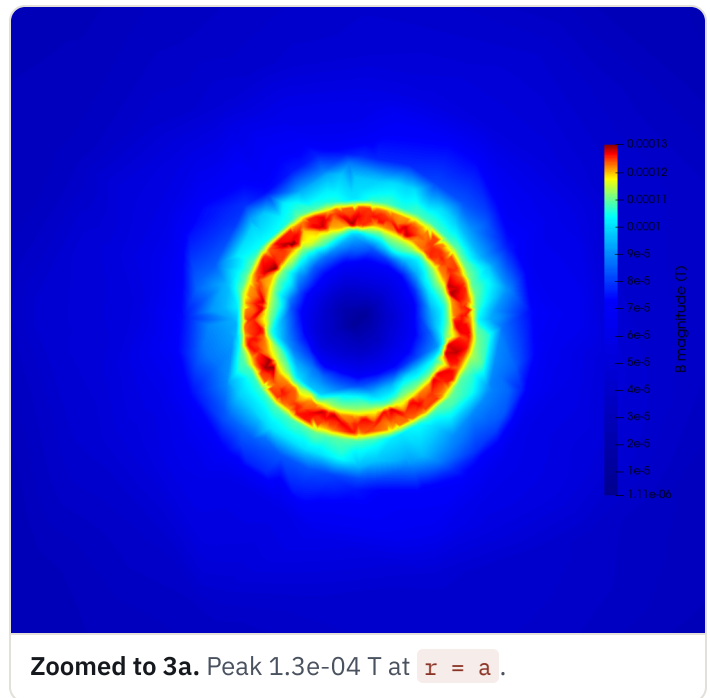
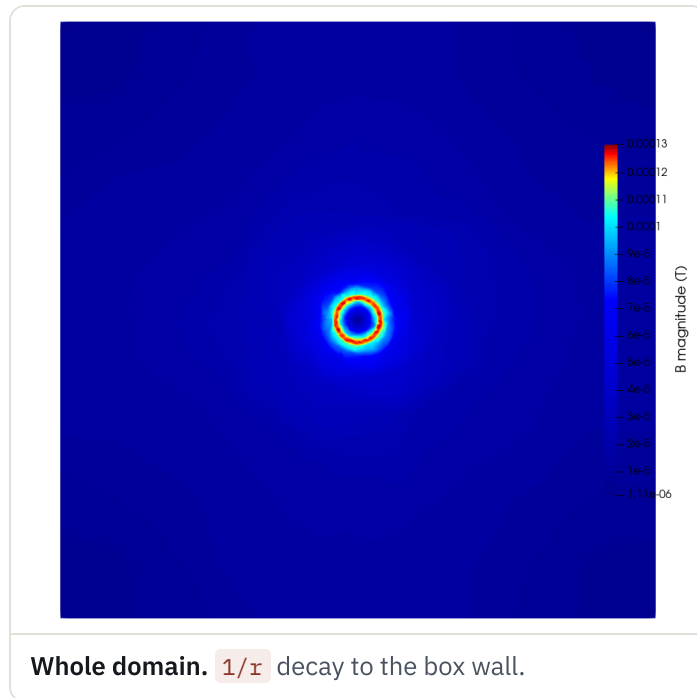


Φ is markedly more linear than in the 1 V case, and on the stricter of two metrics:

	ABSOLUTE	NORMALISED BY V	POINTWISE RELATIVE
1 V case	8.257×10^{-4} V	8.257×10^{-4}	3.946×10^{-3}
1 A case	3.778×10^{-10} V	3.856×10^{-6}	4.301×10^{-6}

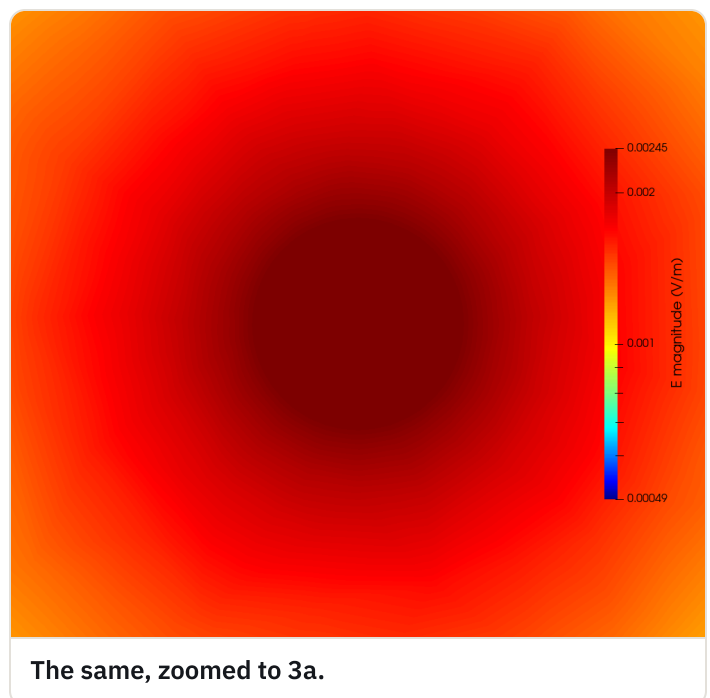
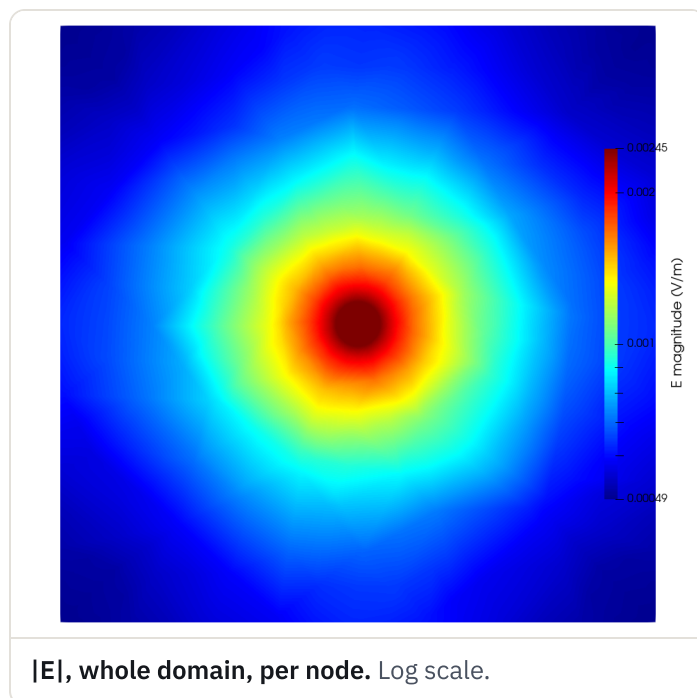
A voltage port forces the whole cap to a single potential; a current port constrains only the total current and lets the cap relax to whatever the field wants. The prescribed equipotential is what bends Φ in the 1 V case — the boundary condition doing its job, not an error.

4. Magnetic flux density



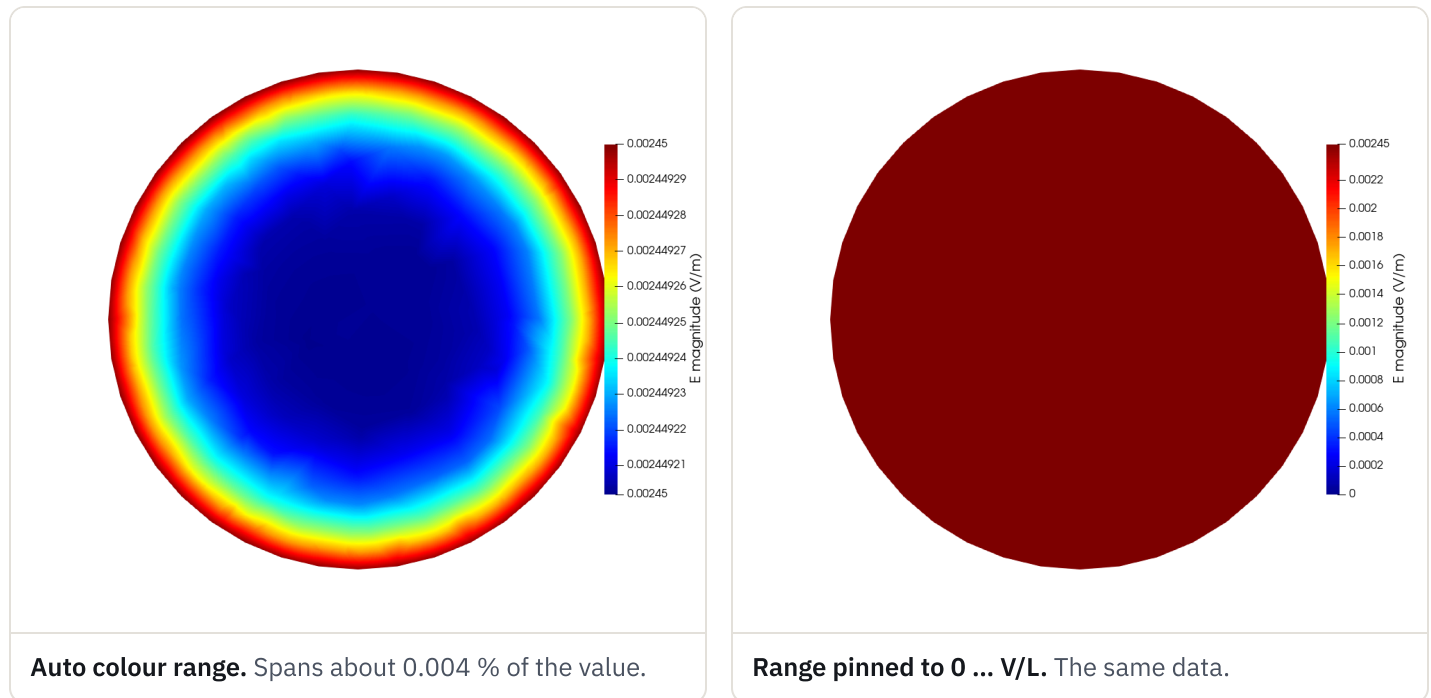
Zero on the axis, rising linearly inside the conductor, falling as $1/r$ outside. The peak of **1.3×10^{-4} T** is the 1 V case's 1.3385 T divided by its 10180 A, which is what driving 1 A instead should give.

5. Electric field



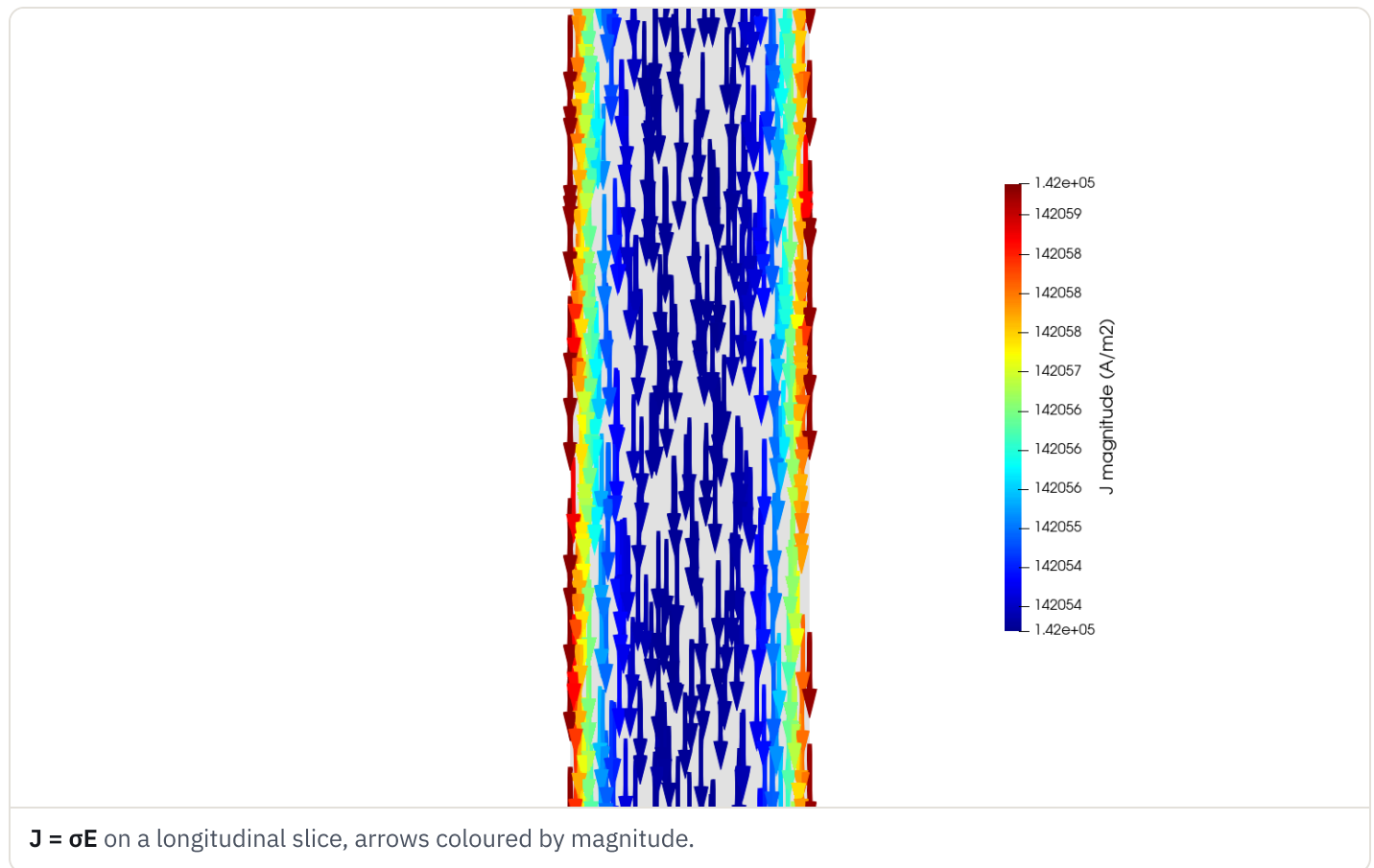
E is one order smoother than B , and structurally so: $E = -j\omega A - \nabla\phi$ is **linear** per tetrahedron because $\nabla\phi$ comes from the P2 space, while $B = \text{curl } A$ is **constant** per tetrahedron.

5.1 Inside the wire, and a warning about colour ranges



E inside the conductor is uniform at $V/L = 2.449\text{e-}03 \text{ V/m}$ — there is no skin effect at $a/\delta = 0.16$. The left figure auto-rescales to the wire's own range, which spans about **0.004 %** of the value, so **noise at the fifth significant figure fills the entire colour map** and reads as a strong radial gradient. The right figure pins the range and the same data renders as the uniform disc it physically is. This is the single easiest way to misread these plots.

6. Current density



The view frames a section of the length rather than all 40 mm. The wire is 3 mm across and 40 mm long, and a view containing the whole thing renders the arrows as a smear two pixels wide — which earlier versions of this figure did. Nothing varies along z , so a section shows everything the full length would.

Read the colours with the §5.1 warning in mind: the bar spans 142,054 to 142,060 A/m², a range of **0.004 %**. $\mathbf{J}$ is uniform; those colours are the fifth digit. The nominal value is $I/A = 1 / 7.0635e-06 = 141,573 \text{ A/m}^2$, and the measured 142,05x sits 0.3 % above it because the current concentrates very slightly toward the surface even at $a/\delta = 0.16$.

7. Validation

Nothing below fits a free parameter.

I collected vs injected	1.000000 A vs 1.0 A — port self-consistency
R	9.796995e-05 Ω against 9.796863e-05 exact — 1.3e-05
L	22.5956 nH — identical to the 1 V case
Φ vs exact z/L , pointwise	4.301e-06
$J(0)/J(a)$	0.999968 against the Bessel 0.999959

All six are asserted automatically by `regression_tests/check.bat`, which runs this case and the 1 V case together. They are duals, and **a change that breaks the duality shows up as the two disagreeing — which neither case alone can see.**

8. Known limitations

- **Azimuthal scatter in B of 1.4–3.1 % near the conductor.** $0(h)$ element noise from B being constant per tetrahedron, the lowest-order quantity in the formulation.
- **Nodal B reads low in the far field**, up to –7.5 % at the box wall against –2 % per cell. The volume-weighted nodal average is dominated by the larger outer elements on a rapidly coarsening mesh.
- **Far-field accuracy was traded for near-field resolution** when the mesh was built: beyond $r = 3 \text{ mm}$ the error is ~2 % rather than ~1 %.
- **Specific to this case: the relative residual is 1.4e-09** where the 1 V case gives 6e-13, while the backward error is as good — 3.8e-21 against 3.7e-21. That is exactly the situation `factorization.hpp` warns about: $\|Ax-b\|/\|b\|$ misleads when $\|b\|$ is small, and a 1 A drive makes a far smaller right-hand side than a 1 V one. **Judge this case by the backward error.**